

Error Analysis

Any measurement is accompanied by uncertainty and error. The measurement error is defined as the difference between the actual value of a quantity and the value obtained by a measurement. In the next parts, we discuss all terms and definitions related to the error analysis.

1 Definitions

The concepts of measurement and error include many terms. Some of these terms are defined as follows:

Resolution – The resolution of a measuring device is defined as the smallest part that can be read or estimated on it. Usually it is a fraction of the smallest division appearing on the measuring device. For example for a ruler in which the smallest divisions is 1 mm, the resolution is about 1 mm. This results in uncertainty in the reading at least by this amount.

Sensitivity – It is the ratio of change in output (or response or scale deflection) of the instrument to change in input or measured variable.

A higher sensitivity indicates that the system can respond to even the smallest input.

Sensitivity has no unique unit. It has wide range of the units which dependent up on the instrument or measuring system

As an example, consider a pressure sensor that has a measurement range of 0–100PSI and an output range of 0–5V. Its sensitivity is .05 Volt/PSI.

Range – is defined as the difference between largest and smallest repeated measurements. The range may increase when we increase number of readings.

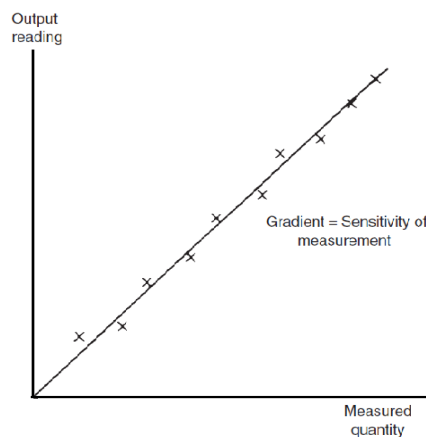
Accuracy – By **Accuracy** we mean how a measurement (or the average value of a set of measurements) might be close to the “true” or the “exact” value. Accuracy is affected mainly by systematic errors explained in the next parts.

Precision – By **Precision**, we mean the degree to which repeated measurements under unchanged conditions are concentrated around their average value. Or simply, the degree to which repeated measurements show the same results as an average (or close to each other). Precision is affected mainly by random errors explained in the [types of errors](#) section.

A measurement system can be precise but not accurate, accurate but not precise, both, or neither. For example, if a measurement contains a systematic error, then increasing the sample size generally increases precision but does not improve accuracy. Eliminating the systematic error improves accuracy but does not change precision.

Repeatability – It describes the closeness of output readings when the same input is applied repetitively over a short period of time, with the same measurement conditions, same instrument and observer, same location, and same conditions of use maintained throughout.

Linearity – It is normally desirable close the output reading of an instrument is linearly proportional to the quantity being measured.



2 Types of errors

In general, errors can be classified into **systematic errors** and **random errors**.

2.1 Systematic errors

Systematic errors are errors that tend to shift all measurements in a systematic way. In that case, their mean value is shifted by the same amount. They occur with the same value shifted for every reading when we use the instrument in the same way and in the same case. Systematic errors can be compensated for if the errors are known. Examples of systematic errors are:

- **Zero error, which** is caused by an incorrect position of the zero point of the scale of the measuring device. For example, when the measuring instrument does not start from exactly zero. Zero errors are consistently present in every reading of a measurement by the same amount and can be positive or negative. Figure 1 shows examples of zero error in a voltmeter (which measures voltage difference in electric circuits).

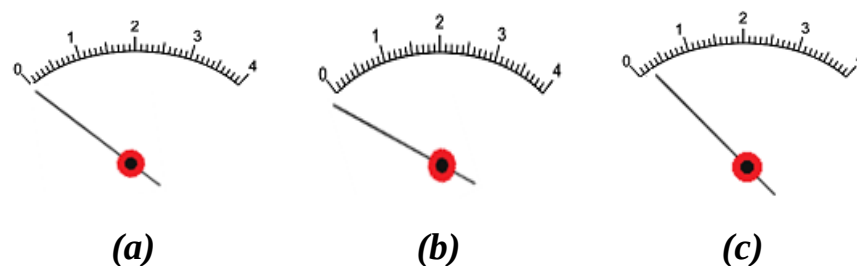


Figure 1: Examples of zero error in a voltmeter.

(a) No Zero Error

The pointer of the voltmeter points to zero when no potential difference exists.

(b) Negative Zero Error

The pointer of the voltmeter does not point to zero but to a negative value when no potential difference exists.

(c) Positive Zero Error

The pointer of the voltmeter does not point to zero but to a positive value when no potential difference exists.

Possible sources of systematic errors are

- An **incorrect calibration** of the measuring instrument.
- Consistent **improper use** of equipment, such as reading an instrument pointer on the wrong scale (personal error).
- The measurement technique “drifts” slightly from one measurement to another

Systematic errors can be reduced by

- Conducting the experiment with care.
- Repeating the experiment by using different instruments.

Control experiments can also be designed to detect the presence of systematic errors. For example, to control for a drift error, the series of measurements might be performed in the opposite order from the original experiment

2.2 Random errors

Random errors are caused by factors that are beyond the control of the observers. They arise from unknown and unpredictable variations in condition and so, they fluctuate from one measurement to the next.

Some sources of random error can be

- **Personal errors** such as human limitations of touch and sight, human reaction speed in time measurements.
- **Low sensitivity** of the instrument.
- **Environmental errors** such as changes in temperature or wind velocity during experiment execution.

Random error can be reduced by:

- Taking repeated readings.
- Calculating the average value of the readings.
- Another example of random error is the parallax error. A parallax error is an error in reading an

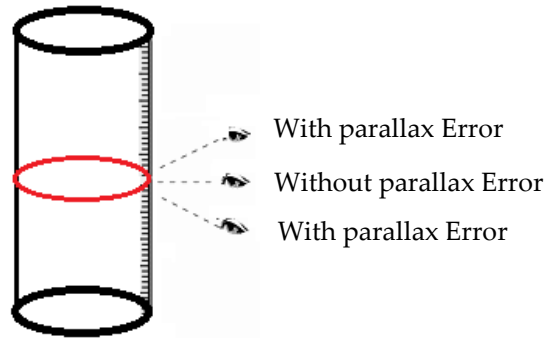


Figure 2: Examples of zero error in a voltmeter.

instrument due to the fact that the eye of the observer and pointer are not in a line perpendicular to the plane of the scale as shown in Figure 2.

As mentioned before, random errors affect the precision of an experiment, while systematic errors can affect its accuracy.

3 “Uncertainty” and “Error” in Probability Theory

Statistical analysis can then be used to estimate the uncertainty of the average value of the readings, i.e. it allows us to quantify random errors. If we take a series of measurements N for a parameter x , the mean average may be more accurate than any individual measurement. We calculate the mean average by,

$$\bar{x} = \frac{\sum_i x_i}{N} \quad (1)$$

The standard deviation is the deviation of each data point from the average of all points. Each data point has a deviation Δx_i , corresponding to the modulus of the difference between the individual measurement and the average,

$$\Delta x_i = |x_i - \bar{x}| \quad (2)$$

If our data is all the population, we define the variance S as the average of the square of the deviations, i.e.

$$S = \frac{\sum_i (\Delta x_i)^2}{N} \quad (3)$$

Notice that we square the magnitude of the deviation so that negative and positive errors won't cancel in our sum.

The standard deviation is defined by,

$$\sigma = \sqrt{S} = \sqrt{\frac{\sum_i (\Delta x_i)^2}{N}} \quad (4)$$

If the data is a sample (a selection taken from a bigger Population), then this sample may not be a good representation of the overall population. Then the variance and the standard deviation in this case are defined by,

$$S = \frac{\sum_i (\Delta x_i)^2}{N - 1} \quad (5)$$

$$\sigma = \sqrt{S} = \sqrt{\frac{\sum_i (\Delta x_i)^2}{N - 1}} \quad (6)$$

Sampling is useful if the population is large. To get information about large population (such as mean and standard deviation), we do not need to look at all members of the population; we only need a sample, but in this case, we lose some precision and possibly some accuracy.

The larger the standard deviation, the more widely spread the data is about the average. For measurements which have only random errors, the standard deviation means that 68% of the measured values are within σ from the average, 95% are within 2σ from average, 99% are within 3σ from average.

In statistical methods, we use the standard deviation σ as the value of the experimental error, and for a set of data points, we write the experimental value of x as,

$$x = \bar{x} \pm \sigma \quad (7)$$

When reporting a measurement we only report the digits that are significant. For an average value, the standard deviation tells us which digit is within the uncertainty limit. We must keep the first digit of the standard deviation, and the corresponding digit in the reported average, but no more. For example, if we measure $x = 0.6231008m \pm 0.00318m$, then we record, $0.623m \pm 0.003m$, because the following digits "-1", "0", "0", and "8" represent values that are less than the uncertainty. Check the appendix for more details on the [significant figures](#).

4 Discussion and Questions

4.1 Show whether the following statements are false or true and state why:

- The precise device is always accurate.
- The precise device has small uncertainty.
- The accurate device has small uncertainty.
- Precision is affected by both systematic and random errors.
- Accuracy is affected by both systematic and random errors.
- Systematic errors can be eliminated by taking many readings and averaging.
- Random errors can be eliminated by good calibration of devices.
- Random errors can be eliminated by taking many readings and averaging.

4.2 When we calculate the variance, we take the average of the square of the magnitude of deviations. Why don't we take the average of the absolute value of the deviations as a measure of the variance?

Appendix

1- More Definitions

Uncertainty - By Uncertainty, we mean how far our result may be from the correct value. Of course, when we are doing an experiment to discover something new, we might have no independent means of determining the correct value! To determine this definition precisely, many concepts are used; the simplest approach is to take the larger of range and resolution as a measure of the uncertainty. For example, suppose that in measuring the width of a non uniform edge block using a cm ruler with a resolution of 0.5 mm (0.05 cm), our measurements in 6 repetitions range from 19.3 cm to 19.6 cm. The uncertainty in this case is taken as the range = 0.3 cm (it is larger than sensitivity).

Uncertainty Ratio (UR) - It is a factor used to compare the experimental result with the reported exact (true) value. It is defined as,

$$UR = \frac{|\text{experimental value} - \text{reported value}|}{\text{Uncertainty}} \quad (8)$$

A value less than 1 indicates good agreement, while values greater than 3 indicate disagreement. Intermediate values need more examination. For example if we measure the gravitational constant as 970 cm/sec² with uncertainty equal to 5 cm/sec², then

$$UR = \frac{|970 - 981|}{5} = 4.2$$

So, in this case we say that the result obtained did not agree with the reported value. But, if the uncertainty in measurements is equal to 30 cm/sec² then:

$$UR = \frac{|970 - 981|}{30} = 0.7$$

then, the result in this case is considered to be in agreement with the reported value.

Significant figures - In a measurement, the significant figures are the number of digits, starting with the first non-zero digit on the left. For example, the measurement $x = 0.043050$ has five significant digits, since we start with the 4 and count the zero in the middle and at the right, while the measurement $x = 0.04305$ has four significant digits.

In multiplication and division, we round the result to the factor or divisor with the smallest significant digits. This can be done either before the multiplication or division, or after. The following examples explain this rule

$$\begin{aligned} 3.5678 \times 0.52 &= 1.855256 = 1.9 \quad (2 \text{ significant digits in } 0.52) \\ 3.5678 \times 0.520 &= 1.855256 = 1.86 \quad (3 \text{ significant digits in } 0.520) \\ 3.5678 / 0.520 &= 6.861154 = 6.86 \quad (3 \text{ significant digits in } 0.520) \end{aligned}$$

When we report our final result then we should round our number of significant digits taking into account the estimated error. We will discuss how to choose the estimated error below.

The Absolute Error - It is the difference between the exact (true) value and measured value. If we don't know the exact value when measuring, we can use the maximum possible error.

The Relative Error - It is the absolute error divided by the exact (true) measurement. If we don't know the actual value when measuring, we can use the measured value instead, i.e.,

$$\text{Relative error} = \frac{\text{Absolute error}}{\text{Exact value(or Measured value)}} \quad (9)$$

The Percentage Error - It is the relative error shown as a percentage.

$$\text{Percentage error} = \frac{\text{Absolute error}}{\text{Exact value(or Measured value)}} \times 100\% \quad (10)$$

2- Propagation of error

Suppose that you have some measurements like $a, b, c, \dots$ with uncertainty in each of these quantities $\sigma_a, \sigma_b, \sigma_c, \dots$. If we need to use the quantities $a, b, c, \dots$ to calculate another quantity, (x), then what is the uncertainty in the calculated value, x ?

The answer may be not very simple due to the propagation of σ_a, σ_b , in the uncertainty of x .

Now, to calculate the uncertainty you have to follow some rules,

2.1 Addition and subtraction

$$x = a + b + \dots + c - g - \dots - h \quad (11)$$

The uncertainty is

$$\sigma_x = \sqrt{(\sigma_a)^2 + (\sigma_b)^2 + \dots + (\sigma_c)^2 + (\sigma_g)^2 + \dots + (\sigma_h)^2} \quad (12)$$

2.2 Multiplication and division

$$x = \frac{ab \dots c}{fg \dots h} \quad (13)$$

The uncertainty formula is

$$\frac{\sigma_x}{x} = \sqrt{\left(\frac{\sigma_a}{a}\right)^2 + \left(\frac{\sigma_b}{b}\right)^2 + \dots + \left(\frac{\sigma_c}{c}\right)^2 + \left(\frac{\sigma_f}{f}\right)^2 + \left(\frac{\sigma_g}{g}\right)^2 + \dots + \left(\frac{\sigma_h}{h}\right)^2} \quad (14)$$

2.3 Multiplication by a constant

$$x = Ka, \quad K = \text{constant} \quad (15)$$

$$\text{The uncertainty will be} \quad \frac{\sigma_x}{|x|} = \frac{\sigma_a}{a} \quad (16)$$

$$\text{Or equivalent to,} \quad \sigma_x = |K| \sigma_a \quad (17)$$

2.4 For the formula of $x=a^n$

$$\frac{\sigma_x}{x} = |n| \frac{\sigma_a}{a} \quad (18)$$

2.5 For the general formula, $x = x(y)$

$$\sigma_x = \left| \frac{dx}{dy} \right| \sigma_y \quad (19)$$

3- The least square regression line

Sometimes we need to get analytically the best line to represent a set of data (x_i, y_i) , where y_i is the dependent variable, and x_i is the independent variable. Then, the equation for the regression line would be,

$$y = a_0 + a_1 x \quad (20)$$

where a_0 is a constant, a_1 is called the regression coefficient. In the least squares regression line (LSRL), the values of a_0 and a_1 are given by,

$$a_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} \quad (21)$$

$$a_0 = \bar{y} - a_1 \bar{x} \quad (22)$$

where $\bar{x}$ and $\bar{y}$ are the average values of x and y , respectively.

Construction of Graphs

1 Construction of graphs on paper

- Sometimes we need to represent results in the form of a graph. Graphs provide a conceptual picture of the information included in the data, and can show qualitative features of the data not clear from just looking at the numbers in the tables. To create a good graph, the following instructions must be taken into consideration:

- 1- Arrange the data to be plotted in a table. Whenever possible, we prefer to plot straight line representations of the data (linear relation). For example, for an object which moves vertically under the effect of the gravity, theoretically, the relation between the distance covered by the object (d) and the time (t) is not a straight line, but the relation between the distance (d) and the time squared (t^2) is a straight line. Therefore, it is preferable to represent this relation between distance and squared time graphically.

An example for the table of the measured data is shown below

Distance by object (d) (m)	Average Time (t) (sec)	t^2 (sec ²)	σ_{t^2}
0.45	0.25	0.063	0.02
0.5	0.4	0.16	0.01
2.8	0.7	0.49	0.04
3.7	0.9	0.81	0.03

- 2- Decide which quantity is to be plotted on the x-axis (the abscissa), and which on the y-axis (the ordinate). Usually the independent variable is placed on the x-axis and the dependent variable on the y-axis. In the last example, we adjust the distance and measure the time. So, distance (d) is the

independent variable on the x-axis, while squared time (t^2) is the dependent variable on the y-axis.

- 3- Decide whether the origin is to appear on the graph. Some uses of graphs require the origin to appear, even though it is not actually part of the data, for example, if the origin is a data point of the plot or if the point at which the “best fit” line intercepts the horizontal or vertical axis is to be determined.
- 4- Choose the scales for the horizontal and vertical axes so that the plotted points representing the data occupy most of the space available for the graph. This can be done by calculating how many units on each axis represent a convenient number of the units of the variable represented on that axis. (In the last example, if the plotting area is 10×10 divisions, then, 10 divisions on x-axis represent 1.0 sec^2 , and 10 divisions in the y-axis represent 4 m). Also, make it easy to locate arbitrary quantities on the graph. (Example: 10 divisions to represent 0.9 sec^2 is a poor choice). Label the major divisions on each axis.
- 5- Write a label in the margin next to each axis, which indicates the quantity being represented and its units.
- 6- Write a label in the margin at the top of the graph that indicates the nature of the graph, and the date when the data was collected. (Example: "Free Fall Object: time² vs. Distance, 23/10/2014").
- 7- Plot each point. The recommended style is a dot surrounded by a small circle. A small cross or plus sign may also be used.
- 8- Draw a smooth curve that comes reasonably close to all of the points. As we mentioned before, whenever possible we plot the data or simple functions of the data so that a straight line is expected. A ruler or other straight-edged object can be used to draw a reasonable fitting straight line. Try to get a similar number of points on each side of the line. Draw a single line all the way across the page as shown in Figure 1 the graph below. Do not simply connect the dots

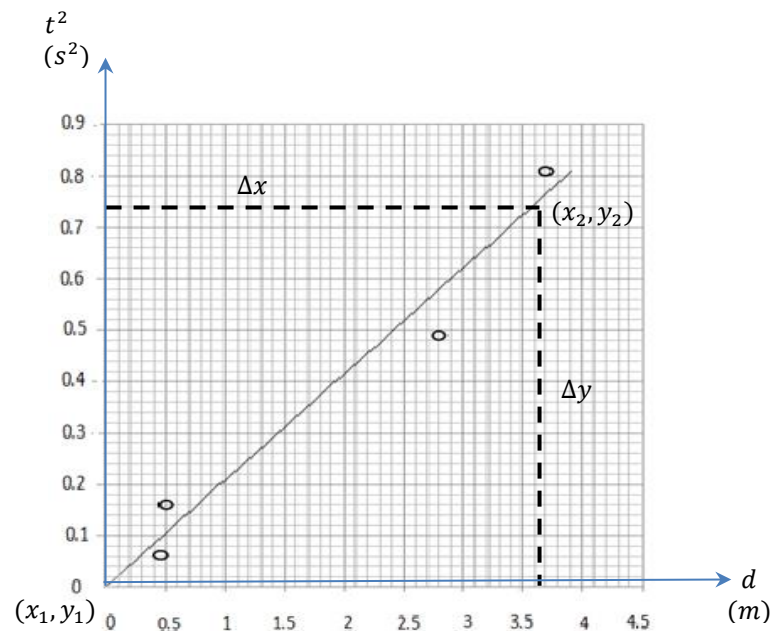


Figure 3: Plotted graph.

- 9- The slope can be estimated by choosing two points on the line and dividing the difference between the two y values by the difference between the two x values. The units of the slope will be the unit of the y axis divided by the unit of the x axis.
- 10- For example in the last plot, pick two points (x_1, y_1) and (x_2, y_2) , as far apart as possible along the “best fit” line and use the slope formula:

$$\text{slope} = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1} \approx \frac{0.70 - 0}{3.40 - 0} = 0.21 \text{ (s}^2/\text{m)}$$

- 11- One can estimate the maximum uncertainty in the slope from the uncertainty at the beginning or end of the measurements, whichever is bigger. Then we divide by the difference between them in x. So, for our example where the final endpoint is $(3.7, 0.81)$, $\sigma_{t^2} = 0.03 \text{ s}^2$, Then, the uncertainty of the slope is given by,

$$\sigma_{\text{slope}} = \frac{0.03}{3.7 - 0.45} = 0.009 \text{ (s}^2/\text{m)}$$

and the slope of the line is reported as follows,

$$\text{slope} = 0.21 + 0.009 (s^2/m) \text{ Construction of graphs using Excel}$$

- The following explanation and figures are based on MS Excel 2013, this is a recommended version due to some added useful features.

- Suppose after performing a certain experiment to verify Newton's second law of motion in the lab, you end up with the following set of data points (which are average values for several trials) that relates the force applied on a certain mass and the acceleration it acquires.

Force (N)	Acceleration (m/s^2)
2.2	9.71
4.3	19.90
6.5	28.56
8.1	35.61

- Open MS Excel 2013 and choose to work on a new blank work book. A blank work book should look like Figure 4. A workbook contains at least one sheet to work on, a default new sheet is created with the new workbook under the name "Sheet 1" seen at the bottom left corner of the work book, other sheets can be added to the workbook by pressing the (+) button next to the sheet name shown in Figure 5.

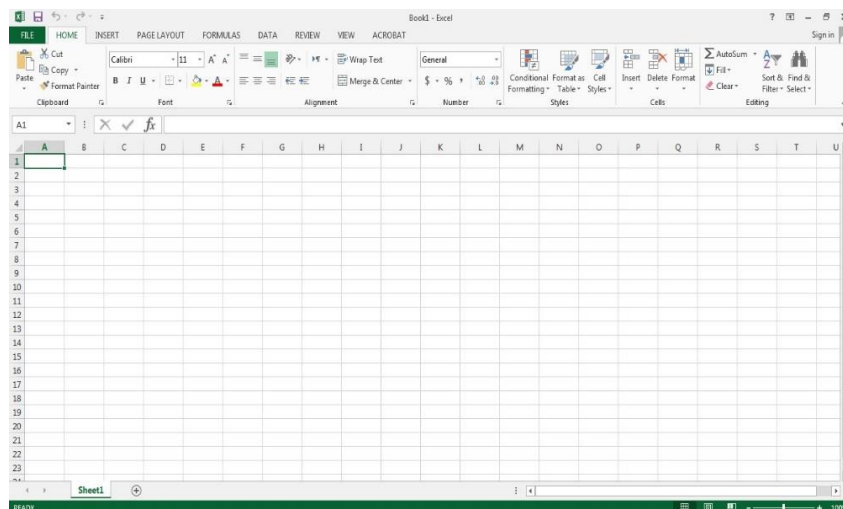


Figure 4: A blank new workbook.

- 3- It is a good practice to rename the sheet with a descriptive name according to the data contained in it. To rename a sheet double click on its name, it will turn into an editable text box and write the new sheet name, “Newton’s 2nd law” in our case.



Figure 5: Sheet names.

- 4- Insert your data in the sheet cells as shown in Figure 6. Always use the row above the data to write the physical quantity the column holds and the unit between brackets and in italics. The thick lines can be added by the selecting the cells then using the border setting (田) button.

	A	B
1	Force (N)	Acceleration (m/s^2)
2	2.2	9.71
3	4.3	19.9
4	6.5	28.56
5	8.1	35.61
6		

Figure 6: inserting the data into the Excel sheet.

- 5- Now, select an empty cell far from the cells containing your data. Go to “Insert” Tab, from the charts area open the “Insert scatter (X,Y) or Bubble chart” to open its drop down list then click on “Scatter” as shown in Figure 7.

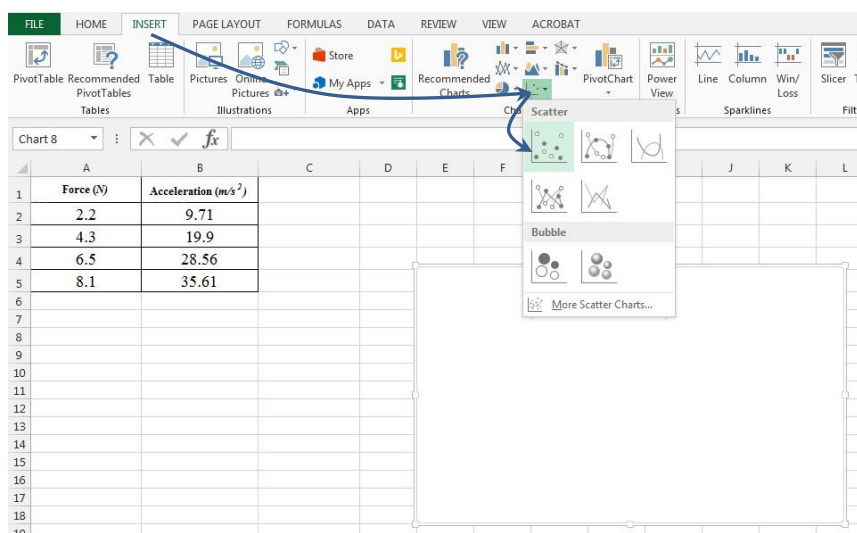


Figure 7: Inserting an empty scatter plot.

- 6- Right click on the empty chart area that appeared after performing the last step and click on “Select Data” shown in Figure 8.

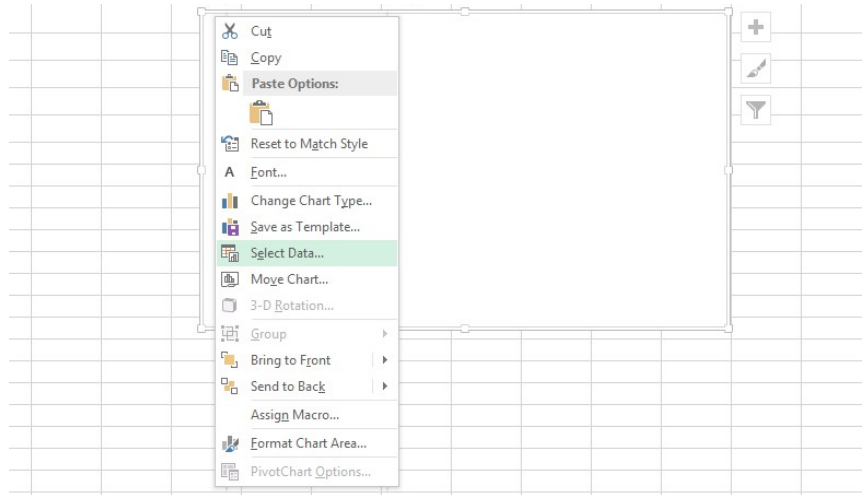


Figure 8: Inserting an empty scatter plot.

- 7- The “Select data source” window opens as shown in Figure 9. This window will allow us to select what to draw on the horizontal axis and what to draw on the vertical. To select the data, click on the “Add” button. In case you have to add more than one plot on the same graph, this “Add” button can be used more than once for each plot.

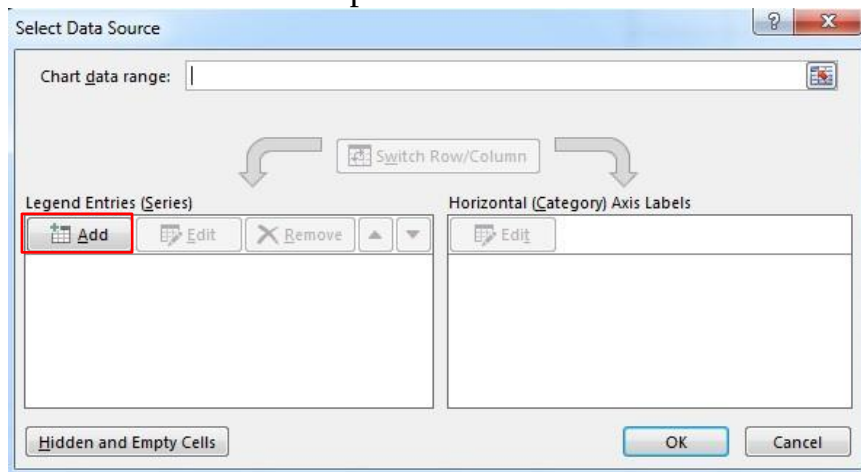


Figure 9: “Select data source” window.

- 8- In the “Edit series” window that will appear, the “series name” field will help us identify which points describe what in case we have more than one plot on the same graph. We will enter the series name as a *descriptive* title to the plot “The force applied on a certain mass and the acceleration it acquires” (the date when the data was collected may also be added). For the “Series X values” field, we will choose the values to add on the x-axis, just click on the text box below “Series X values” and delete any thing written in it then select the force values using your mouse as shown in Figure 10 (Don’t select the title cell “Force (N)”), the text box is now filled automatically. Do the same for the “Series Y values” to choose the vertical axis values. The “Edit series” window after filling it will look like Figure 11. Click “Ok” to close this window and also “Ok” in the “Select data source” window to close it.

Force (N)
2.2
4.3
6.5
8.1

Figure 10: Selecting the x-values.

Figure 11 shows the "Edit Series" dialog box. The "Series name" field contains the text "The force applied on a certain mass and the acceleration it acquires". The "Series X values" field contains the formula "= 'Newton's 2nd law'!\$A\$2:\$A\$5". The "Series Y values" field contains the formula "= 'Newton's 2nd law'!\$B\$2:\$B\$5". The "OK" and "Cancel" buttons are visible at the bottom.

Figure 11: “Edit series” window after filling it.

- 9- The chart area will now look as in Figure 12, with the data points on the graph. As you can see, the series name appeared as a title above the graph.

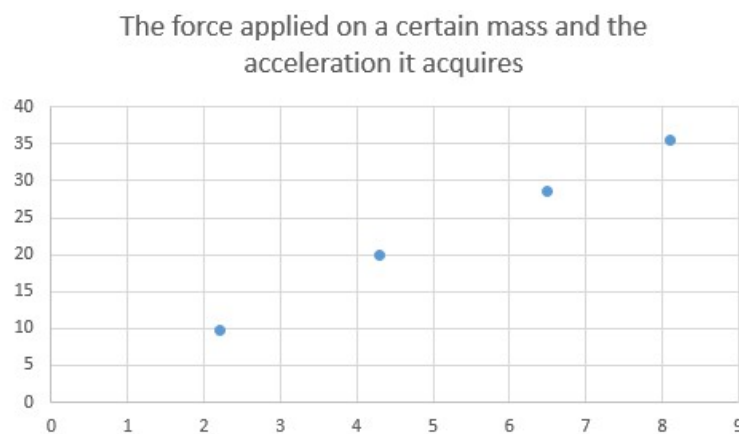


Figure 12: Data points on the graph.

To add the best fitted line passing through these points and describing the relation, click on one of the data points then right click and select “Add Trendline” as shown in Figure 13. A dotted line called “Trendline” will appear passing through the data points as shown in Figure 14.

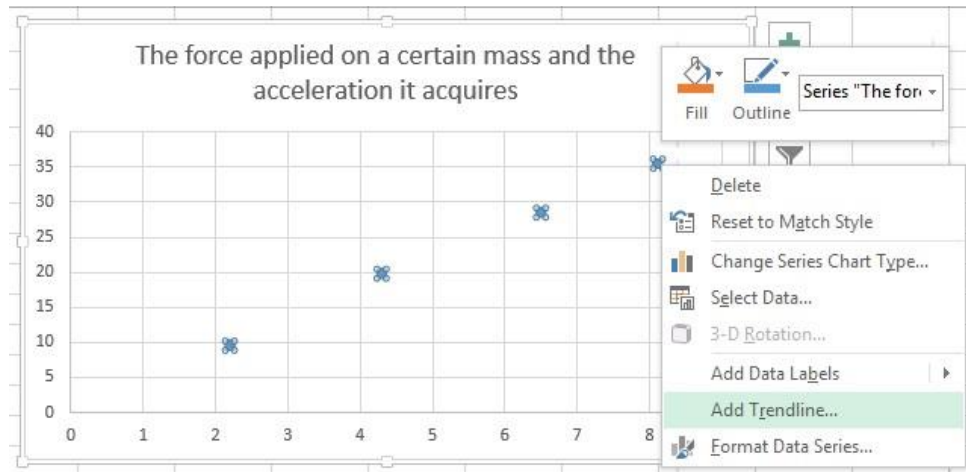


Figure 13: Adding a Trendline.

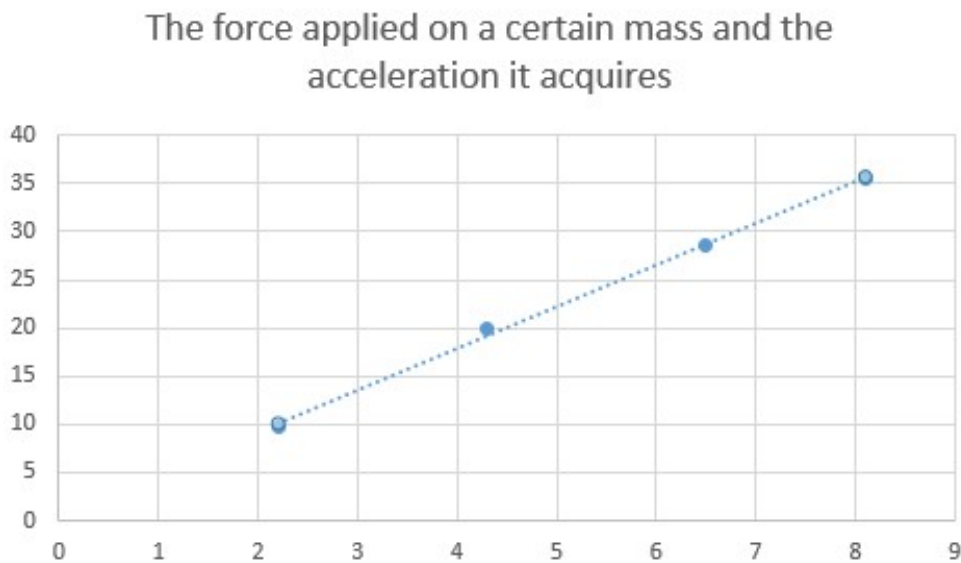


Figure 14: A Trendline passing through the data points.

10- The Trendline still needs some adjustments, you will notice that the “Format Trendline” options will appear after the last step on the right of the screen as shown in Figure 15. The options of our interest include the type of the Trendline, in our case it will be “Linear” because the relation between the force and the acceleration in Newton’s 2nd law is described by a straight line. The “Set intercept” is useful in case we know that the straight line should cut a certain point on the vertical axis. In our case, we know that when the force applied on a mass is zero, its acceleration is also zero so we must set the intercept to “0.0”.

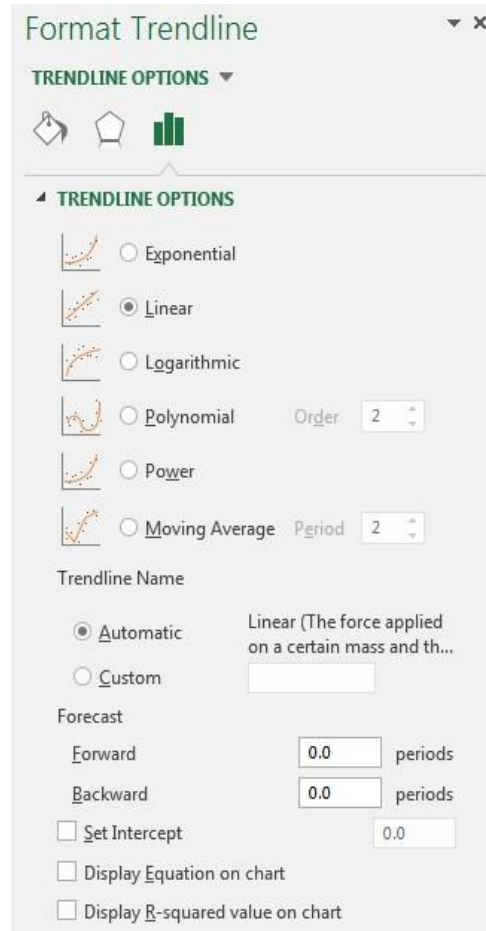


Figure 15: Trendline options.



Setting the intercept to “0.0” will not cause the Trendline to be extended until it passes through the origin because the Trendline is drawn from the minimum x-axis value to the maximum x-axis value (from 2.2 to 8.1 in our case). However, the extension of the Trendline will certainly be passing through the intercept point.

The next important element is “Display equation on chart” which writes the equation that describes the drawn Trendline itself (the linear best fitted line). In our example we can make use of the slope of the drawn Trendline as will be shown later.

The “Display R-squared value on chart” displays a value called R-squared value (also called coefficient of determination) which is a statistical measure of how close the data points fits the drawn Trendline. It might be very useful in choosing the type of the Trendline in case we do not know whether the relation is linear, exponential, second order, etc... The R^2 value ranges from 0 to 1. With $R^2 = 1$ the Trendline actually passes through all the data points.



You can always re-open these Trendline options by selecting the Trendline on the graph and right click then click “Format Trendline”.

The Trendline equation and the R-squared value are now visible on the chart as shown in Figure 16.

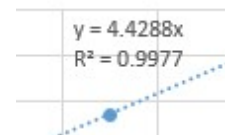


Figure 16:
Trendline data.

- 11- We are now close to finalizing our graph but it is still missing some important elements. Select the chart by clicking on it. A plus sign button will appear on its top right corner. On clicking on this plus button a set of check boxes appears, the important check boxes to be checked are shown in Figure 17. You now need to edit the axis titles, just double click on each of them and rename. The best practice is to write the physical quantity and its unit between brackets and in italic as shown in Figure 18.

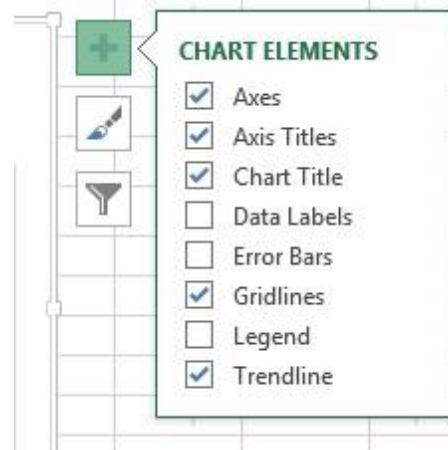


Figure 17: Chart elements.

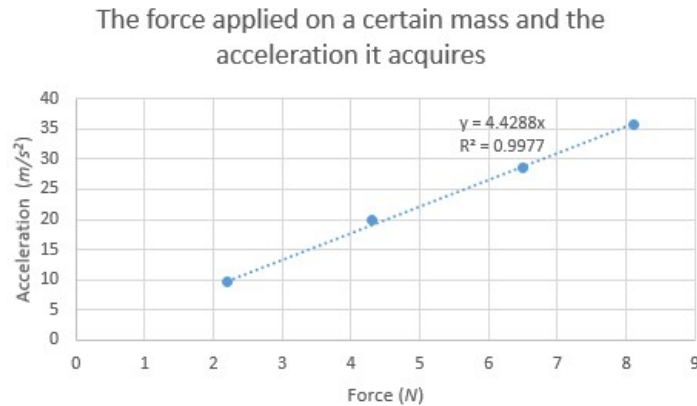


Figure 18: The chart after renaming the axes titles.

- 12- You can make use of the Trendline equation to find the value of the mass m , recall that the force was plotted on the x -axis and the acceleration is plotted on the y -axis, so the Trendline equation $y = 4.4288x$ describes the relation $a = 4.4288 F$ with $m = \frac{1}{4.4288}$.

Adding error bars to the plot

- Error bars are a graphical representation of the uncertainty introduced in the measurements. They can be used for one of the quantities drawn on any of the axis or both quantities (i.e Force and mass in the example above).
- 1- Assume that the experiment described above is repeated several times for error analysis and the *standard deviation* for each quantity was calculated and inserted in a table as shown.

Force (N)	Acceleration (m/s ²)	σ_F	σ_a
2.2	9.71	0.13	0.19
4.3	19.90	0.23	0.63
6.5	28.56	0.45	1.14
8.1	35.61	0.32	1.42

2- Add the last two columns in the table above containing the errors for each quantity to the Excel sheet.

3- To add these errors as error bars on the previously added plot, click on the graph to select it, click on the plus sign that appears at the top right corner then put the cursor on the arrow besides the “Error bars” check box and select “More Options...” as shown in Figure 19.

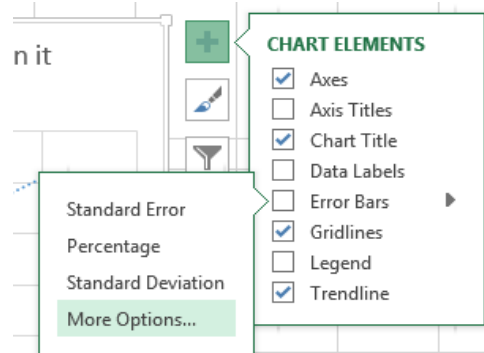


Figure 19: Opening the “Error bars” options.

4- The error bars appear on each point on the graph as shown in Figure 20. Also, a right pane appears for the “Error bars” options.

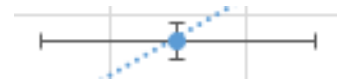


Figure 20: Error bars on a graph point.



The vertical bars represent the uncertainty in the quantity draw on the vertical axis (i.e acceleration). The horizontal bars represent the uncertainty in the quantity draw on the horizontal axis (i.e force).

5- To set the vertical bars length to describe the uncertainty $\pm \sigma_a$, select the vertical bars by clicking on any of them on any point. The selected bars should look like Figure 21.



Figure 21: Selecting the vertical error bars.

6- The right pane now has the options for the selected vertical error bars as shown Figure 22, the “Direction” allows adding positive error bars only $+\sigma_a$, negative error bars only $-\sigma_a$ or both $\pm \sigma_a$.

- 7- To set the error bars to the calculated values in the table above, Select “Custom” for the “Error amount” option then click on “Specify Values”. A “Custom error bars” window appears with two text boxes for the positive and negative values for the error bars, in our case we have both values equal $\pm \sigma_a$. Delete any text from the text boxes and select the values for the standard deviation in the measured acceleration. The “Custom error bars” window will look like Figure 23 after choosing the correct set of data.
- 8- Repeat the previous three steps for the horizontal error bars representing the uncertainty in measuring the force σ_F . The final plot after adding the error bars will be as shown in Figure 24.



Figure 22: Error bars options.

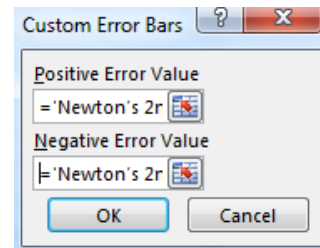


Figure 23: Custom Error bars window.

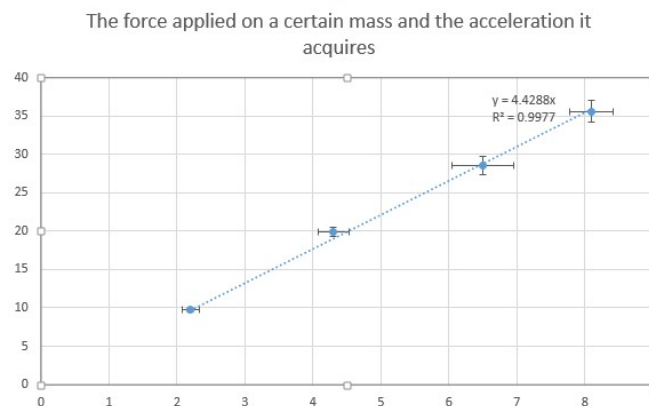


Figure 24: The chart after adding error bars.